

AI PLACEMENT COPILOT

Multi-Agent Resume, Job Matching & Interview Preparation

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1. Introduction

- Placement preparation is fragmented across resumes, job portals, and interview practice.
- A resume may contain skill gaps or miss role-specific keywords.
- Job listings are often mismatched with a candidate's skills, projects, and preferences.
- Interview preparation is usually generic and lacks structured feedback.
- The proposed solution is a multi-agent copilot that connects all three activities.
- One coordinator routes tasks and produces a single, actionable preparation plan.

2. Problem Statement & Objectives

Problem Statement

Placement preparation is fragmented. Candidates struggle with resume gaps, mismatched job listings, and weak interview practice.

Objectives

- Critique and tailor the resume to a target role
- Rank internships and jobs against the candidate profile
- Run mock interviews with structured feedback
- Coordinate agents using shared candidate context
- Generate one consolidated preparation plan

3. Proposed Multi-Agent Pipeline



Core idea: one coordinator routes the candidate through specialized agents and combines their outputs.

4. System Architecture

Candidate Input — Resume, skills, target role

Coordinator Agent — Plans and routes tasks

Resume Agent — Critiques and tailors resume

Matcher Agent — Scores jobs and internships

Interview Agent + Output — Mock Q&A, feedback, action plan

5. Agent Workflow & Responsibilities

Resume Agent

Find missing skills, weak bullets, ATS gaps, and tailor content to the target role.

Matcher Agent

Compare profile features with job requirements and rank opportunities by fit.

Interview Agent

Generate role-specific questions, evaluate answers, and give structured feedback.

Shared Context

Maintain candidate skills, projects, target role, preferences, and previous feedback.

Coordinator Output

Combine agent results into priorities, next actions, and a placement preparation plan.

6. Technologies Used

Python

Core backend and data processing

n8n

Workflow automation and agent orchestration

LLM / AI Models

Reasoning, rewriting, matching, and interview generation

NLP

Resume parsing, skill extraction, and keyword analysis

Job Database / APIs

Internship and job listing retrieval

Vector Database

Semantic similarity between profile and job descriptions

Telegram / Web UI

Candidate interaction and result delivery

JSON / REST APIs

Structured data exchange between agents

7. Technical Challenges & Solutions

Fragmented candidate data

Normalize resume, skills, projects, preferences, and target role into one profile.

Resume gaps

Use role requirements to identify missing skills and suggest targeted improvements.

Job mismatch

Use weighted skill, keyword, and semantic scoring to rank relevant openings.

Generic interviews

Generate questions from the target role and candidate profile.

Agent coordination

Use a coordinator with structured inputs/outputs and shared context.

Actionable feedback

Convert scores and comments into prioritized next steps.

8. Applications & Future Scope

Applications

- Resume optimization for a specific role
- Internship and job discovery
- Personalized interview preparation
- Placement-readiness tracking
- Student career guidance

Future Scope

- Continuous resume improvement from interview feedback
- Live job-market and company matching
- Adaptive interview difficulty
- Skill-gap learning recommendations
- Application tracking and reminders
- Personalized career roadmap

9. Conclusion

A multi-agent Placement Copilot turns fragmented preparation into one coordinated workflow.

The Resume Agent improves role alignment, the Matcher Agent prioritizes relevant opportunities, and the Interview Agent builds interview readiness.

The Coordinator connects these outputs and produces a consolidated plan with clear next actions.

THANK YOU